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Section 3.2: Ways in which Data is Stored and Read

In this section we will look at the 2 main ways in which data is accessed, stored and read:

- **Serial Access** (also known as 'sequential access')
- **Direct Access** (also known as 'random access')

Key Concepts of this section:

- # Know the difference between '**Serial**' and '**Direct**' data access.
- # Understand how both **serial** and **direct access** work.
- # Be able to **state** the **different uses** of serial and direct data access storage methods.

Serial Access

Overview:

- # This method is also sometimes called '**Sequential Access**'.
- # Data is accessed by **starting at the beginning** and then **searched through**, in **order/sequence**, until the required information is found.

Kind of like an old film reel where the movie starts at the beginning and data is read in order until the movie is finished

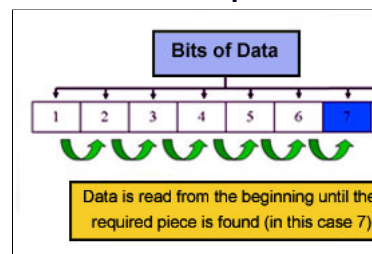
- # The need to search from the beginning of the storage medium makes this type of data access **very slow**.



Key Words:

Serial, Sequential, Sequence, Magnetic Tape

Examples:



Serial data access reads data in 'sequence'

Where Serial Access is used:

- # Serial access is used on **old magnetic tapes**.
- These are used where it is necessary to store a lot of data but where speed of access is unimportant.

Magnetic Tapes are still used for **backing up** large amounts of data in organisations like **schools** and **business**.



- # Serial access medium (like magnetic tapes) are also used in **batch processing systems** like payroll and in the preparation of utility bills.

The way that serial access works means that **all of the data has to be read** and there is **no danger** of **missing out** someone's wage payment or bill invoice.

Remember: Serial access works by accessing the data at the beginning and then working through it bit-by-bit until the end.

This is why there is no danger of missing anything out during the batch processing examples above.

- # Examples of storage medium that use serial access are:

- **Video cassettes**
- **Music cassettes**



Old film reels access the movie's scenes in 'sequence' from start to finish.



Magnetic tape can store a lot of data but is slow to access it.

- **Backup tape cartridges**(hold lots of tape for backing up large amounts of data).

Direct Access

Overview:

- # This method is also sometimes called '**Random Access**'.
- # The computer can **calculate exactly** where the **data has been stored** and can go **straight to it** directly (instant access).

The computer knows where all of the data on the storage medium is and so it can **access** it **very quickly**.

It is not forced to start accessing the data at the beginning and then working through bit-by-bit like with serial access.

A good comparison as to how direct access differs to serial access is to think of how **DVD movies** work **compared** to old **film-reel movies**.

In a DVD movie you can '**jump**' **directly** to **any scene** you want but in a film-reel you have to **move through every scene** until you come to one you want.

- # This 'direct' method of accessing data is **much faster** than serial access.

Where Direct Access is used:

- # This method is used with storage medium such as:

- **Hard Disk Drives**
- **CD's and DVD's**

Direct access is used on most modern storage devices where speed of access is important.

Note: The large size of modern files makes direct access even more useful.



Hard Disk Drive

- # Direct access medium are used in situations where fast access to data is important like in **online booking systems** and **point-of-sale (POS)** terminals in shops.

For example: You wouldn't want to wait around at a shop checkout while the system was trying to access the price of beans!

- # Other examples of storage medium that use direct access are:

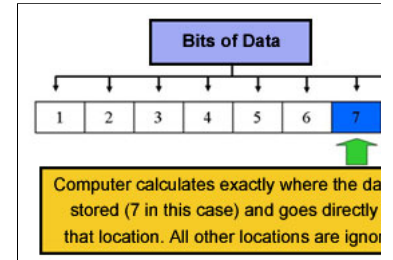
- **USB memory sticks**
- **Flash memory** (like in digital cameras and mobile phones)
- **Blu-ray discs**



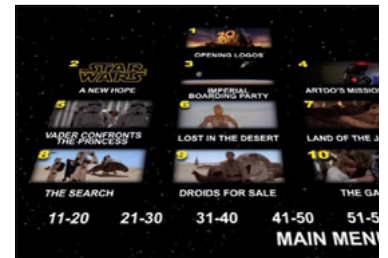
Key Words:

Direct, Random

Examples:



Direct access can go straight to the required data 'directly'.



DVD movies allow you to jump to scenes. This is possible through direct access.



Flash memory cards are an example of storage that use direct access.



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